REVIEW ARTICLE

Systematic review of liver directed therapy for uveal melanoma hepatic metastases

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Abstract

Background: Uveal melanoma (UM) is a rare malignancy with a propensity for metastasis to the liver. Systemic chemotherapy is typically ineffective in these patients with liver metastases and overall survival is poor. There are no evidence-based guidelines for management of UM liver metastases. The aim of this study was to review the evidence for management of UM liver metastases.

Methods: A systematic review of English literature publications was conducted across Ovid Medline, Ovid MEDLINE and Cochrane CENTRAL databases until April 2019. The primary outcome was overall survival, with disease free survival as a secondary outcome.

Results: 55 studies were included in the study, with 2446 patients treated overall. The majority of these studies were retrospective, with 17 of 55 including comparative data. Treatment modalities included surgery, isolated hepatic perfusion (IHP), hepatic artery infusion (HAI), transarterial chemoembolization (TACE), selective internal radiotherapy (SIRT) and Immunoembolization (IE). Survival varied greatly between treatments and between studies using the same treatments. Both surgery and liver-directed treatments were shown to have benefit in selected patients.

Conclusion: Predominantly retrospective and uncontrolled studies suggest that surgery and locoregional techniques may prolong survival. Substantial variability in patient selection and study design makes comparison of data and formulation of recommendations challenging.

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Background

Uveal melanoma (UM) is a rare malignancy that arises from the uveal tract of the eye. It is biologically distinct from cutaneous melanoma (CM), with differences in clinical behaviour, response to treatment and mutation profile.[1,2] Recent decades have yielded a dramatic increase in the understanding of the biology of UM, in particular the cytogenetic and signal transduction pathways contributing to carcinogenesis and metastatic potential.[2] Despite these advances survival from UM remains largely unchanged, with no change in five year survival during the period from 1973 to 2008.[3] A key clinical determinant of survival from UM is the development of liver metastases. The liver is the most common site of metastasis from UM, affected in 70–90% of cases, and is the only site of metastasis in approximately 50% of cases.[4–8] Survival following development of liver metastases is typically short, with median survival of approximately two to three months.[6,8,9]

Evidence-based guidelines regarding the management of UM were published in 2015.[10] To date there is no consensus regarding management of liver metastases from UM. Systemic chemotherapy traditionally used in CM has poor response rates when used in metastatic UM.[11] Various locoregional treatment strategies have been described, including surgery, isolated hepatic perfusion, hepatic artery infusion of chemotherapy, transarterial chemoembolisation, selective internal radiotherapy and Immunoembolization.

Management of liver metastases from UM is largely experiential, with most studies from large centers being retrospective in nature. There are no guidelines on best practice in this area, and there remains a lack of consensus on how to manage this rare condition. Therefore, the aim of this study was to undertake a systematic review of the evidence for management of liver metastases from UM, comparing treatments in terms of overall survival and disease-free survival.

Methods

Search strategy

The study protocol was designed prospectively and registered online with PROSPERO, the international prospective register of systematic reviews. Literature searches were conducted up to the 10th of April 2019 in three electronic databases: Ovid Medline, Ovid EMBASE and Cochrane CENTRAL. Date of coverage was not restricted and results were limited to articles in English only. In Ovid Medline and EMBASE, the search strategies used a combination of subject headings (MeSH and Emtree) and keywords, following pilot testing. Subject headings included "Liver neoplasms" qualified with subheading "sc" (secondary) which was then combined with the exploded heading "Uveal neoplasms". Keywords included liver with the proximity operator "adj5" metasta* which was then combined with uvea* "adj5" melanoma*. In Cochrane CENTRAL, keyword combinations were used. The complete search strategies for Ovid Medline is outlined in Appendix 1. Search strings were adapted for use across the other two databases.

Inclusion and exclusion criteria

All English-language studies that included patients with liver metastases from UM, with treated primary disease and the liver as the only site of metastasis, were considered. Given the likely paucity of high-quality studies both clinical trials and observational studies were included. Case reports, non-published data and conference abstracts were excluded. The interventions examined were systemic therapy, as well as targeted liver-only therapy including, but not limited to: surgery, ablation, isolated hepatic perfusion (IHP), hepatic artery infusion (HAI), transarterial chemoembolization (TACE), selective internal radiotherapy (SIRT) and Immunoembolization (IE). The primary outcome was overall survival from the time of treatment, with disease-free survival as a secondary endpoint.

Data extraction

All abstracts and full text articles were reviewed independently by two authors (AR and BK) and considered for eligibility. Disagreement regarding eligibility was resolved by consensus. A single author (AR) was responsible for data extraction, which for each study included study design, study population, details of the intervention described as well as any comparison treatment, outcomes measured and survival data. Treatments were categorized into groups for qualitative analysis, with results summarized within each treatment type (surgery, IHP/percutaneous hepatic perfusion (PHP), HAI, TACE, SIRT, IE, other). Due to the heterogeneity of the data, meta-analysis was not feasible or appropriate.

Results

The initial search yielded 727 articles after removing duplicates (Fig. 1). A total of 55 studies were included in the final analysis. Included studies were categorized according to the predominant intervention that was used in the study. These studies included 2446 patients altogether, with 39 studies being retrospective cohort studies. There were two randomized controlled trials and 14 prospective cohort studies.

Surgery

Studies examining the use of surgical resection are summarized in Table 1. A total of 10 studies were included. These studies were predominantly retrospective analyses of surgical cohorts, with either no comparison group or comparison with historical controls or best supportive care. A total of 793 patients were treated with surgery, with median overall survival (OS) ranging from 10 to 35 months. OS was higher for those patients undergoing surgery than those undergoing systemic chemotherapy or best supportive care (9–15 months).[4,12,13] The largest retrospective study of 255 patients reported a median OS post resection of 14 months, compared with 8 months for patients without surgery.[14]

Several surgical techniques were described, with surgical resection alone being the most common intervention. Resection was combined with intra-arterial chemotherapy or HAI in two studies (75 patients),[13,15] with radiofrequency ablation (RFA) in two studies (29 patients)[12,16] and with TACE in one study (three patients).[17] The safety of laparoscopic RFA and laparoscopic segmental resection was reported in 16 patients, with no operative mortality and 19% operative morbidity.[12] In that study median OS was 35 vs. 15 months ($p < 0.001$) for those undergoing resection or systemic chemotherapy, respectively. No major liver resections were performed in the surgery group. One study compared 57 patients undergoing surgery alone with 13 patients undergoing surgery and intraoperative RFA of unresectable tumours.[16] The mean size of ablated tumours was 13 mm, with no recurrences reported at the site of ablation over a median follow up of 63 months. OS was 27 vs. 28 months for the surgery group and combined surgery/RFA group respectively, suggesting that RFA may be a useful adjunct for local control with unresectable disease.

IHP/PHP

The technique of IHP has been described by Alexander et al.[18] Briefly, IHP involves vascular isolation of the liver and

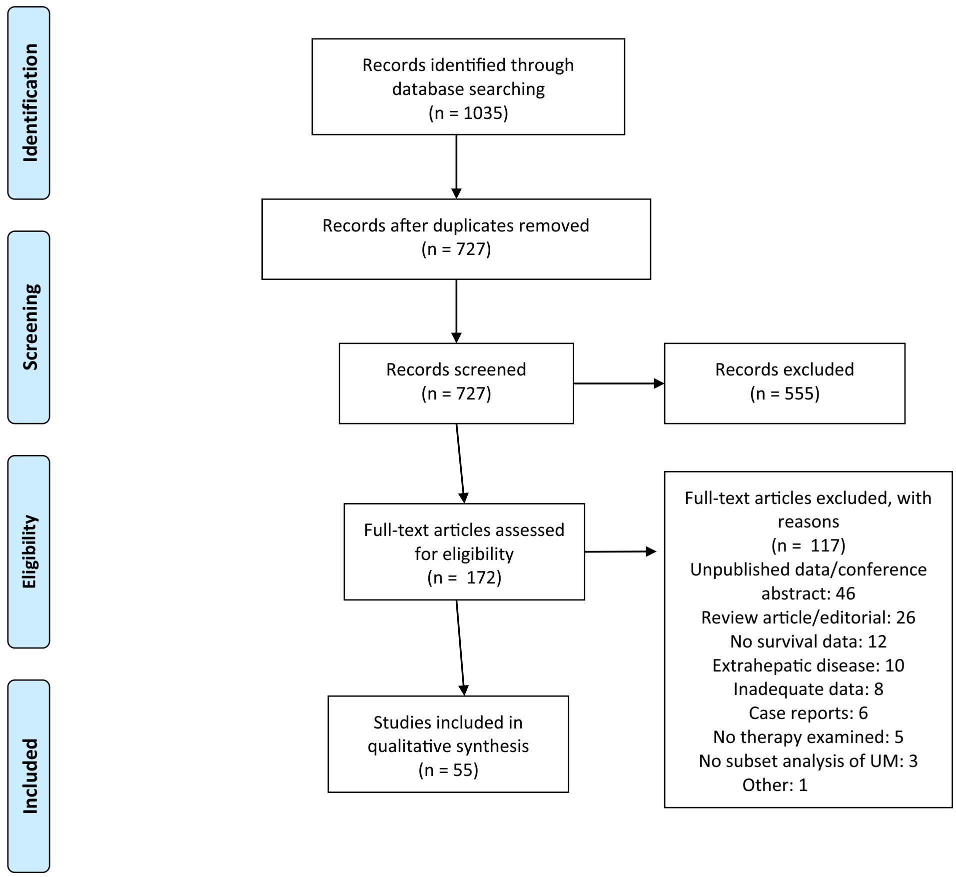

Figure 1 Selection and inclusion flow diagram

perfusion with heated chemotherapy using an extracorporeal bypass circuit for 60 min. Ten studies utilized either IHP or percutaneous IHP (PHP) to treat UM liver metastases (Table 2). Of these, three studies involved prospective cohorts, with the remainder being retrospective analyses. There were no randomized studies in this group. A total of 293 patients were treated with these techniques, with reported median OS ranging from 9 to 25 months. The predominant chemotherapeutic agent used was Melphalan, which has a high degree of hepatic extraction. Melphalan was used at a variety of doses, and in two studies was compared with Melphalan with perfusate buffering[19] and tumour necrosis factor-alpha (TNF-alpha).[20] Several studies reported substantial morbidity and mortality from the IHP procedure. The largest cohort of 61 patients noted a 30-day mortality of 7%, with this rate reducing to 2% with more recent institutional experience.[21] Median operating time was >8 h, and median estimated blood loss was 2–3.5 L.[20,22] PHP as a minimally invasive alternative to IHP has been reported for 18 patients.[23] The procedure was well tolerated in most patients, with no procedure-related mortality, and a partial response rate of 44%. The main advantage of PHP over IHP is that it can be repeated in patients with a partial response or stable disease post-therapy. Median OS was 10 months. Similar results were described in a further retrospective analysis of 51 patients from two centres, with a median OS of 15 months.[24]

HAI

There were eight studies evaluating the use of HAI in UM liver metastases (Table 3). Four of these studies were prospective, with one randomized controlled trial (RCT). A total of 405 patients were treated, with reported OS ranging from 10 to 24 months. Several chemotherapeutic agents were used, predominantly Fotemustine, but also including Carboplatin, Melphalan and Cisplatin/Vinblastine/Dacarbazine combination therapy. The technique of HAI was achieved through either a subcutaneously implanted port or femoral artery catheter access. Leyvraz and colleagues conducted a RCT of 171 patients comparing HAI Fotemustine with IV Fotemustine.[25] They found no difference in OS between HAI (15 months) and IV Fotemustine (14 months). However, HAI was associated with a longer PFS than IV therapy (4.5 vs. 3.7 months, HR 0.62, $p = 0.002$), although this failed to translate into longer OS.

TACE

Seventeen studies evaluated the use of TACE in UM liver metastasis (Table 4). Of these, five were prospective cohort

Table 1 Studies including surgery for management of UM liver metastases

Author	Design	n	Intervention	Comparison	Median survival
Salmon[15] (1998)	PC	75	Resection and HAI	Historical controls	10 mos with surgery, 9 mos overall
Rietschel[4] (2005)	RC	93	Resection	No treatment, chemo	32 mos with surgery, 9 mos with systemic/no treatment
Rivoire[13] (2005)	RC	63	Resection+- HAI	Chemo, supportive care	15 mos overall, 25 mos with R0 resection
Frenkel[42] (2009)	RC	74	Resection	No surgery	23 mos for surgery, 7 mos without
Mariani[14] (2009)	RC	255	Resection	No surgery	14 mos with surgery vs 8 mos without
Marshall[43] (2013)	PC	12	Resection	Nil	24 mos with surgery (11 out of 12 had systemic relapse), all liver mets 12 mos.
Yang[17] (2013)	RC	8	Resection, TACE (+- PEI or RFA)	Nil	11 mos with resection, 7 mos with other treatment
Gomez[44] (2014)	RC	97	Resection	No surgery	27 mos with surgery/ablation, 8 mos without
Akyuz[12] (2016)	RC	44	Laparoscopic resection or lap. RFA	Systemic therapy	35 mos with surgery, 15 mos with systemic therapy
Mariani[16] (2016)	RC	72	Resection	Resection + RFA	27 mos with surgery vs 28 mos surgery + RFA

PC, prospective cohort; RC, retrospective cohort; DFS, disease free survival; OS, overall survival; PEI, percutaneous ethanol injection.

Table 2 Studies of IHP or PHP for treatment of UM liver metastases

Author	Design	n	Intervention	Comparison	Median survival
Alexander[20] (2000)	PC phase I-II	22	IHP	IHP with TNF	9 mos. 5% mortality from procedure
Noter[22] (2004)	RC	8	IHP	Nil	10mos. PFS 7 mos.
van Iersel[45] (2008)	RC	13	IHP	Nil	10 mos. PFS 7 mos. Hepatic PFS 8 mos.
Olofsson[46] (2014)	PC phase II	34	IHP	Nil	24 mos.
van Iersel[47] (2014)	PC phase I	3	IHP	Nil	19 mos.
Ben-Shabat[21] (2016)	RC	61	IHP	Nil	22 mos.
de Leede[48] (2016)	RC	31	IHP	Nil	10 mos. PFS 6 mos.
Ben-Shabat[19] (2017)	RC	52	IHP	IHP, perfusate buffering	25 mos. PFS 10 mos.
Vogl[23] (2017)	RC	18	PHP	Nil	10 mos.
Karydis[24] (2018)	RC	51	PHP	Nil	15 mos. PFS 8 mos. Hepatic PFS 9 mos.

PC, prospective cohort; RC, retrospective cohort; TNF, tumour necrosis factor; CR, complete response; PR, partial response; PFS, progression free survival.

studies, incorporating a total of 647 patients. The reported OS ranged from 5 to 29 months. A wide variety of chemotherapeutic agents and treatment protocols were used, with many patients receiving multiple TACE treatments and other treatment modalities. The largest prospective study of TACE included 30 patients, with a median of three TACE treatments administered to 24 patients, and six enrolled patients not treated due to progressive disease before TACE was administered.[26] The response rate amongst the treated patients was 20%, with a median OS of 5 months. Survival was related to degree of response, with OS extended to 22 months in patients with complete or partial response, and 9 months in those with stable disease post treatment. Eight patients developed grade 3 or 4 complications, with three patients developing vascular thromboses following TACE that prevented further treatment.

SIRT

Six studies evaluated the use of SIRT for UM liver metastases, all of which are retrospective analyses (Table 5). This technique utilizes a beta-emitting yttrium radioisotope that is delivered to the liver via an intra-arterial catheter, and its use in UM was first described in 2009.[27] Treatment populations varied significantly in terms of pre-treatment prior to SIRT, both locoregional and systemic, as well as hepatic burden of disease. Reported median OS ranged from 9 to 24 months.

Table 3 Studies of HAI for treatment of UM liver metastases

Author	Design	n	Intervention	Comparison	Median survival
Leyvraz[49] (1992)	Pilot	14	HAI fotemustine	Nil	12 mos.
Egerer[50] (2001)	PC phase I	7	HAI fotemustine	Nil	24 mos.
Becker[51] (2002)	PC phase II	23	HAI fotemustine, IL-2 and TNF-alpha	IV fotemustine	HAI survival 369 days vs 349 days with IV chemo
Peters[52] (2006)	RC	101	HAI fotemustine	Nil	15 mos.
Melichar[53] (2009)	RC	10	HAI cisplatin/vinblastine/dacarbazine	Nil	16 mos.
Farolfi[54] (2011)	RC	18	HAI fotemustine, carboplatin	Nil	21 mos.
Heusner[55] (2011)	RC	61	HAI melphalan or combination	Nil	10 mos.
Leyvraz[25] (2014)	RCT	171	HAI fotemustine	IV fotemustine	15 mos. Longer PFS with HAI 4.5 vs 3.7 months HR 0.62, p = 0.002

PC, prospective cohort; RC, retrospective cohort; RCT, randomized controlled trial; IL-2, interleukin-2; TNF, tumour necrosis factor; PFS, progression free survival.

Table 4 Studies of TACE for treatment of UM liver metastases

Author	Design	n	Intervention	Median survival
Maglivit[56] (1988)	RC	30	TACE with cisplatin	11 mos.
Bedikian[11] (1995)	RC	201	TACE, HAI, BE	TACE 6 mos. vs. chemo 5 mos.
Morassut[57] (1995)	RC	9	TACE with cisplatin/DCZ	16 mos.
Patel[26] (2005)	PC phase II	30	TACE with BCNU	5 mos.
Vogl[58] (2007)	PC	12	TACE with mitomycin C	21 mos.
Dayani[59] (2009)	RC	21	TACE with cisplatin, doxorubicin and mitomycin	8 mos.
Fiorentini[60] (2009)	Pilot	10	TACE with DEBIRI	8 of 11 patients alive at median 6 mos follow up
Huppert[61] (2010)	Pilot	14	TACE with cisplatin/carboplatin	11 mos. PFS 8 mos.
Schuster[62] (2010)	RC	25	TACE fotemustine or cisplatin	5 mos.
Edelhauser[63] (2012)	RC	21	TACE with fotemustine	29 mos. PFS 7.3 mos.
Venturini[64] (2012)	PC phase II	5	TACE DEBIRI	All patients alive at follow up 8–13 mos.
Farshid[65] (2013)	RC	20	TACE with mitomycin C	20 mos.
Duran[66] (2014)	RC	15	TACE with doxorubicin, mitomycin C	6 mos.
Carling[67] (2015)	RC	14	TACE with DEBIRI	TACE 15 mos vs chemo 7 mos.
Gonsalves[68] (2015)	RC	50	TACE with BCNU	7 mos.
Valpione[69] (2015)	RC	141	TACE with CPT-11	15 mos.
Shibayama[70] (2017)	RC	29	TACE with cisplatin	23 mos.

RC, retrospective cohort; PC, prospective cohort; BE, bland embolization; DCZ, dacarbazine; BCNU, 1,3-bis(2-chloroethyl)-1-nitrosourea; ITT, intention to treat; DEBIRI, drug-eluting beads with Irinotecan; PFS, progression free survival; OS, overall survival.

Other

The remaining studies of therapy for UM liver metastases are listed in Table 6. These include three studies of Immunoembolization (IE) with granulocyte-macrophage colony stimulating factor (GM-CSF). Valsecchi et al. conducted randomized trial comparing IE with bland embolization in 53 patients, with median OS of 21 months vs. 17 months for IE and bland embolization, respectively.[28] The difference in survival was significant for those patients with more than 20% liver involvement (p = 0.047) but not in other groups. Of note, all patients developed progressive liver disease, highlighting the often palliative nature of treatment for UM liver metastases. Both IE and bland embolization were well tolerated, with no reported mortality or delayed morbidity. LASER-induced interstitial thermotherapy (LITT), a novel ablative technique, has been utilised for UM liver metastases.[29] LITT was delivered to 18 patients with 44 tumours under MRI guidance, with a reported median survival of 3 years. Details of pre-treatment of these patients was not reported. LITT was well tolerated, with no mortality or major morbidity reported.

Table 5 Studies of SIRT for treatment of UM liver metastases

Author	Design	n	Intervention	Comparison	Median survival
Kennedy[27] (2009)	RC	11	SIRT	Nil	No median survival available. 1 YS 80%
Gonsalves[71] (2011)	RC	32	SIRT	Nil	9 mos.
Klingenstein[72] (2013)	RC	13	SIRT	Nil	19 mos.
Schelhorn[73] (2015)	RC	8	SIRT	Nil	20 mos.
Eldredge-Hindy[74] (2016)	RC	71	SIRT	Nil	24 mos.
Xing[75] (2017)	RC	15	SIRT	Nil	11 mos.

RC, retrospective cohort, 1 YS, one year survival.

Table 6 Studies of other treatments of UM liver metastases

Author	Design	n	Intervention	Comparison	Median survival
Sato[76] (2008)	PC phase I	34	IE	Nil	14 mos. PFS-Liver 5 mos. PFS-Systemic 10 mos.
Yamamoto[77] (2009)	RC	53	IE	CE with BCNU	IE 20 mos. vs. 10 mos. with CE p = 0.005
Eichler[29] (2014)	RC	18	LITT	Nil	3 years
Valsecchi[28] (2015)	RCT	53	IE	BE	IE 21 mos. vs. 17 mos. with BE

GM-CSF, granulocyte-macrophage colony-stimulating factor; PFS, progression free survival; CE, chemo-embolisation; BCNU, 1,3-bis(2-chloroethyl)-1-nitrosourea; LITT, LASER-induced interstitial thermotherapy; OS, overall survival; BE, bland embolisation.

Discussion

This is a systematic review of the management of liver metastases from UM, incorporating data from 55 studies and 2446 patients. The published literature varied considerably in terms of patient selection, disease extent, study design and outcome measurement. Despite the large number of studies included in this review, there were few prospective studies, and even fewer with comparative data. As a result, formulating clear management guidelines for this condition is challenging. However it remains important to consider how these patients are managed. Despite our improved understanding of the biology of the disease, survival once liver metastasis has occurred remains limited. Results with systemic chemotherapy have been poor, and response rates to immunotherapy are in the order of 5–10%.[30–34] Patients with liver metastases from UM require safe and efficacious palliative treatment options.

Surgical resection of UM liver metastasis has been described by a number of authors in this review, with comparable safety to liver resection for other indications.[35] This series suggests that median OS is prolonged in surgical patients, in the range of 8–72 months. These findings are tempered by considerable selection bias, in that those with resectable disease likely have a lower burden of disease and potentially more favourable tumour biology compared with non-resectable patients. Whilst it is difficult to make firm recommendations based on low quality evidence, liver resection can be considered in carefully selected patients with good life expectancy, technically resectable disease and no extrahepatic metastases. This approach is similar to that described for liver resection for other non-colorectal, non-neuroendocrine liver metastases (NCNNLM).[36–38] Combining surgical resection with RFA may also present an alternative for local control with unresectable disease.[12,16]

A number of hepatic arterial based therapies have also been used for UM, relying on the principle of predominantly arterial-based blood supply to metastatic tumours of the liver. With the exception of IHP, most were well tolerated with low morbidity and mortality. Determining the ideal therapy for those patients with unresectable disease is difficult, considering the heterogeneous nature of the studies examined. Retrospective analyses of experience with IHP/PHP, HAI, TACE, SIRT and IE have all suggested a survival benefit in selected patients. The decision of the treating clinician as to which therapy is used is likely to depend on patient factors, as well as the experience and techniques available at the treating institution.

The dilemma remains as to how to select those patients with favourable tumour biology. In their retrospective review of 255 surgical patients, Mariani and colleagues reported that a disease-free interval of more than 24 months correlated with improved survival (11% two year survival vs 39%, $p < 0.001$)[14]. This suggests that simple clinical data may predict who will benefit most from liver resection for metastatic UM. In addition, biological markers of the primary tumour such as monosomy 3 and lack of tumour suppressor gene BAP1 expression have been linked to increased propensity for metastasis and poorer survival, but to date these have not been evaluated for their influence on treatment choice once metastatic UM presents.[2,39,40] While it is beyond the scope of this review to explore prognostic factors for UM, future studies may consider how these factors influence therapy chosen for UM liver metastases.

This review has been limited by the low quality of evidence available, which has been noted in a previous analysis.[41] The low incidence of UM makes accrual into prospective studies slow and impractical. In addition, the variability in burden of disease and inherent selection bias in choosing treatment makes comparing treatment groups difficult. Designing a study of treatment groups with meaningful comparison of efficacy between them is a challenging proposition. Future studies must endeavor to use similar selection criteria, treatment protocols and endpoints to enable comparison of outcomes.

Management of liver metastasis from UM remains a significant challenge to the clinician. Predominantly retrospective and uncontrolled studies have suggested that both surgery and locoregional techniques may prolong survival. Given that there is no prospect of cure for these patients, any proposed treatment must be well tolerated and offer symptom and local disease control. Further research is required to delineate the relative merits of these therapies.

Appendix 1. Medline search strategy

Search strategy: MEDLINE (OVID).

1 exp Liver Neoplasms/sc [Secondary]
2 exp Uveal Neoplasms/
3 exp melanoma/
4 2 and 3 (6251)
5 1 and 4 (338)
6 (liver adj5 metasta*).mp.
7 (uvea* adj5 melanoma*).mp.
8 6 and 7
9 5 or 8
10 exp animals/not humans.sh.
11 9 not 10
12 limit 11 to English language

Conflicts of interest

None declared.

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